MILITARY SPECIFICATION

TEST SET MK 339 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of Test Set Mk 339 Mod 0, Bureau of Naval Weapons LD 476761. Test Set Mk 339 Mod 0 simulates the input and output conditions of an electrically set torpedo system for checking the operation of torpedo servo amplifiers used in all mods of Amplifier Mk 140. Test Set Mk 339 Mod 0 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, or such portions thereof as designated herein, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment Class HI (High Impact) Shipboard Application, Tests for
MIL-T-945	Test-Equipment, For Use with Electronic Equipment; General Specification
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18404	Torpedoes, Environmental Requirements; General Specification for

FSC 4931

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MIL-F-18870 — Fire Control, Optical, and Associated Equipment Intended for Shipboard Use; the General Specification for

MIL-W-21927 — Weapons, Handling and Preparation for Delivery of, General Requirements for

STANDARDS

MILITARY

MIL-STD-108 — Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-129 — Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

BUWEPS LD 476761 — Test Set Mk 339 Mod 0, List of Drawings, Subassemblies, Parts, and Specifications

BUWEPS DWG 1349424 — Container, Shipping - Test Set Mk 339 Mod 0

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

OP 2555 — Amplifier Mk 140 Mods 0, 1, and 2, and Test Set Mk 339 Mod 0, Description, Operation, and Maintenance

OD 12406(NAVORD) — Test Set Mk 339 Mod 0, Factory Acceptance Test

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description. - Test Set Mk 339 Mod 0 simulates input and output conditions of an electrically set torpedo system. It is used while servicing and maintaining the servo amplifiers used in all mods of Amplifier Mk 140. A cable with a printed circuit connector connects the test set to a servo amplifier. The test set is operative when the cover is removed and the power plug is connected to a 115-volt, 400-cycle, 3-phase power supply. Switching is provided so that the incoming line voltage can be indicated by the output meter. An adjustment control is provided so that the line voltage can be regulated until 115 volts is indicated on the meter. For a complete description of Test Set Mk 339 Mod 0 and information pertinent to all phases of its operation and use, refer to OP 2555.

3.2 General requirements

3.2.1 Preproduction sample. - When specifically required by the contract or order, preproduction samples of the test set shall be furnished. The sample will be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of Test Set Mk 339 Mod 0 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUWEPS LD 476761.

3.2.3 Finishes, coatings, construction, and dimensions. - Finishes, coatings, construction, and dimensions for Test Set Mk 339 Mod 0 shall be strictly in accordance with applicable drawings, specifications, and publications listed in BUWEPS LD 476761.

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3.2.4 Moisture, Fungus, and Corrosion. - Test Set Mk 339 Mod 0 shall be rendered moisture, fungus and corrosion resistant in accordance with detailed requirements specified in the drawings and specifications listed in BUWEPS LD 476761.

3.2.5 Marking. - Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings listed in BUWEPS LD 476761.

3.2.6 Interchangeability. - Unless otherwise specified on the applicable drawings, component parts of assembled units fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior revisions and correction of the applicable drawings (see 3.2.10).

3.2.7 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies shall meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government shall furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.8 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.2.9 Requests for deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

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3.2.10 Conflicting requirements. - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definititions

3.3.1 Vibration. - Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude refers to displacement, which is given in inches peak to peak. The frequency is given in cycles per second (cps).

3.3.2 Shock. - Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build up of 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration (g's), duration in milliseconds (ms) and some description of the shock wave form.

3.3.3 Inclination. - Inclination as refered in this specification is defined as the acute angle between a given plane and a true vertical plane.

3.3.4 Tropical environment. - Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which present an opportunity for fungi growth.

3.4 Performance requirements and product characteristics. - Test Set Mk 339 Mod 0 shall meet the following performance requirements and product characteristics:

3.4.1 Performance requirements

3.4.1.1 Dielectric. - The wiring of Test Set Mk 339 Mod 0 shall show no evidence of damage during and after the dielectric tests specified in 4.4.4.1, Dielectric.

3.4.1.2 Insulation resistance. - The wiring insulation of Test Set Mk 339 Mod 0 shall show a minimum resistance of three megohms and no evidence of damage during and after the test specified in 4.4.4.2, Insulation resistance.

3.4.1.3 Ground, damping signal, and power circuit continuity. - Circuit continuity in Test Set Mk 339 Mod 0 shall be as specified in 4.4.4.3, Ground, damping signal, and power circuit continuity.

3.4.1.4 Phase Rotation lamps. - When Test Set Mk 339 Mod 0 is connected and tested as specified in 4.4.4.4, the Phase Rotation lamps shall light to indicate proper phase rotation.

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3.4.1.5 Signal input dials. - When Test Set Mk 339 Mod 0 is connected ıd tested as specified in 4.4.4.5, the AC voltmeter shall indicate the ɔllowing readings when the Signal Input dials are set as specified.

Signal Input 1 Dial Settings	AC Voltage Readings
0.50 volt	0.50 ± 0.03 volt
1.00 volt	1.00 ± 0.05 volt
2.50 volts	2.50 ± 0.13 volt
5.00 volts	5.00 ± 0.25 volt
7.50 volts	7.50 ± 0.38 volt
9.99 volts	9.99 ± 0.50 volt

Signal Input 2 Dial Settings	AC Voltmeter Readings
0.50 volt	0.50 ± 0.03 volt
1.00 volt	1.00 ± 0.05 volt
2.50 volts	2.50 ± 0.13 volt
5.00 volts	5.00 ± 0.25 volt
7.50 volts	7.50 ± 0.38 volt
9.99 volts	9.99 ± 0.50 volt

3.4.1.6 Output meter and Output Meter switch. - When Test Set Mk 339 lod 0 is connected and tested as specified in 4.4.4.6, the output meter shall ndicate the following readings when the Output Meter switch is set as specified:

	Output Meter Readings
Output Meter switch set at AMP:	2.5 ± .38 volt
	6.0 ± .56 volt
	11.5 ± 1.0 volt
Output Meter switch set at NOISE:	0.5 ± .06 volt
	0.8 ± .06 volt
	1.0 ± .06 volt

3.4.1.7 Program Selector switch, decks D, E, F, and G. - When Test ɜet Mk 339 Mod 0 is connected and tested as specified in 4.4.4.7, the Program ɜelector switch shall give AC voltage indications of 0, 4, or 8 volts and DC voltage indications of 0 or approximately 24 volts.

3.4.1.8 Program Selector switch, deck A, and Signal Direction switch. - When Test Set Mk 339 Mod 0 is connected and tested as specified in 4.4.4.8, the Program Selector switch, deck A, and the Signal Direction switch shall perform the switching operation as specified therein.

3.4.1.9 Program Selector switch, decks B and C, and Phase Output lamps.- When Test Set Mk 339 Mod 0 is connected and tested as specified in 4.4.4.9, the Program Selector switch and Phase Output lamps shall operate as specified therein.

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3.4.1.10 Tuning eye synchronism indicator. - When Test Set Mk 339 Mod 0 is connected and tested as specified in 4.4.4.10, the tuning eye synchronism indicator shall be fully open at 0 volt and fully closed at 100 volts ± 15 volts.

3.4.1.11 DC power supply output. - When Test Set Mk 339 Mod 0 is connected and tested as specified in 4.4.4.11, the internal DC power supply shall have an output voltage of 24 ± 2 volts DC.

3.4.2 Product characteristics

3.4.2.1 Power requirements. - Test Set Mk 339 Mod 0 shall be energized from a 115-volt (± 7 volts), 400-cycle (± 2-cycles), 3-phase, 200-watt (minimum power source.

3.4.2.2 Degree of enclosure. - Test Set Mk 339 Mod 0 shall be enclosed in a spraytight case in accordance with the requirements specified in MIL-STD-108, Spraytight.

3.4.2.3 Simulated synchro outputs. - Test Set Mk 339 Mod 0 shall be capable of producing simulated 400-cycle synchronous torpedo function orders of gyro angle, running depth, and enabling run. The function orders shall be reversible to simulate an increasing or decreasing function.

3.4.2.3.1 Gyro angle orders. - Test Set Mk 339 Mod 0 shall be capable of producing simulated gyro angle orders in single- and double-speed modes, and "zero return" orders in single- or double-speed modes.

3.4.2.3.2 Running depth orders. - Test Set Mk 339 Mod 0 shall be capable of producing simulated running depth orders in single-speed high, single-speed low, and double-speed modes, and "limits of operation exceeded" orders for each mode.

3.4.2.3.3 Enabling run orders. - Test Set Mk 339 Mod 0 shall be capable of producing simulated enabling run orders in single-speed mode, and "limits of operation exceeded" orders.

3.4.2.4 Phase rotation indicator. - Test Set Mk 339 Mod 0 shall be capable of indicating the phase rotation of phases ABC and CBA of the incoming power source.

3.4.2.5 Output phase indicator. - Test Set Mk 339 Mod 0 shall be capable of indicating the output phase of a servo amplifier as being "in" or "out" of phase with the incoming power source.

3.4.2.6 Output meter. - Test Set Mk 339 Mod 0 shall incorporate one AC voltmeter capable of indicating the line voltage, signal output, and noise level voltage of a servo amplifier.

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3.4.2.7 Synchronism indicator. - An electron ray tube shall be incorporated in Test Set Mk 339 Mod 0 to simulate the function of a tuning eye synchronism indicator.

3.4.2.8 Test jacks. - Test Set Mk 339 Mod 0 shall have test jacks to permit observation by oscilloscope of a servo amplifier damping signal wave form without interaction of amplifier input signals.

3.4.2.9 Controls and switches. - All controls and switches necessary to operate Test Set Mk 339 Mod 0 shall be located on the face plate.

3.4.2.10 Fuses. - Circuitry of Test Set Mk 339 Mod 0 shall be adequately fuse protected.

3.4.2.11 Cable and connector stowage. - Cables and connectors designed for use with the test set shall be stowed within the instrument case when not in use.

3.4.2.12 Cover. - The cover of Test Set Mk 339 Mod 0 shall be gasketed.

3.5 Environmental requirements

3.5.1 Vibration. - Test Set Mk 339 Mod 0 shall withstand vibration requirements as follows:

3.5.1.1 Vibration, transportation, storage, and handling. - Test Set Mk 339 Mod 0, packed and packaged as defined in section 5, shall withstand the tests specified in 4.5.1.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.5.1.2 Vibration, operating condition. - Test Set Mk 339 Mod 0 shall withstand the tests specified in 4.5.1.2, and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.2 Shock. - Test Set Mk 339 Mod 0 shall withstand shock requirements as follows:

3.5.2.1 Shock, transportation, storage, and handling. - Test Set Mk 339 Mod 0, packed and packaged as defined in section 5, shall withstand the tests specified in 4.5.2.1, and thereafter shall show no damage and shall meet all the requirements of section 3.

3.5.2.2 Shock, operating condition. - Test Set Mk 339 Mod 0, mounted in accordance with 4.5.2.2.1, with circuits energized, shall withstand the tests specified in 4.5.2.2, and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

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3.5.3 Inclination. - Test Set Mk 339 Mod 0, mounted in accordance with 4.5.3.1, shall withstand the inclination tests specified in 4.5.3.2, and during and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.4 Temperature and humidity. - Test Set Mk 339 Mod 0 shall withstand the temperature and humidity conditions specified in 4.5.4, and thereafter shall show no corrosion or damage and shall meet all performance requirements and product characteristics of this specification.

3.5.5 Tropical environment. - Test Set Mk 339 Mod 0 shall withstand the fungus resistance tests specified in 4.5.5, and thereafter shall show no fungus growth, corrosion, or other damage, and shall meet all performance requirements and product characteristics of this specification.

3.5.6 Marine atmosphere environment. - Test Set Mk 339 Mod 0 shall withstand the simulated sea atmosphere conditions and tests specified in 4.5.6, and thereafter shall show no evidence of corrosion or damage and shall meet all performance requirements and product characteristics of this specification.

3.5.7 Operating life. - Test Set Mk 339 Mod 0 shall be subjected to the accelerated life test specified in 4.5.7.

3.6 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. - The instrument, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assembly, painting, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.7.1 In addition to the examination and tests performed on the preproduction samples, the workmanship exhibited therein will be evaluated to determine acceptability. The approved standard of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of three (3) Test Sets Mk 339 Mod 0, shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of Test Set Mk 339 Mod 0 by the contractor prior to approval of the contracting activity or the completion of tests on preproduction samples shall be at the contractor's risk. Pre-production samples accepted shall be applied as part of the quantity specified by the contract or order.

4.1.2 Periodic Production Samples. - The tests of 4.4 and 4.5 shall be evaluated as follows:

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory compliance of each unit from (3) three consecutive groups, select one (1) unit from each group of twenty-five (25).

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.

4.2 Classification of tests. - The inspection and testing of Test Set Mk 339 Mod 0 shall be classified as follows:

4.2.1 Acceptance tests. - These tests are to be accomplished on each test set being submitted for acceptance under contract. These tests are defined in 4.4 and shall be conducted in accordance with instructions in NAVORD OD 12406. Failure in any of these tests will result in the rejection of the test set.

4.2.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.1 and 4.1.2 which are representative of the production of Test Set Mk 339 Mod 0 after the award of the contract to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.5. Failure of a sample in any of these tests will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

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4.3 Test equipment. - The following items of test equipment are required to perform the tests set forth in this specification:

4.3.1 Voltage breakdown tester. - Model P-2 (Industrial Instruments), or equivalent.

4.3.2 Megohmer. - Model B-5 (Herman H. Sticht Co.), or equivalent.

4.3.3 Vacuum-tube voltmeter. - Model 400 D (Hewlett Packard), or equivalent.

4.3.4 Volt-Ohm-Milliammeter. - Model 260 (Simpson), or equivalent.

4.3.5 Oscilloscope. - Model 208B (Dumont), or equivalent.

4.3.6 Variac. - 0- 130 volts, 400 cycles, 2.5 ampere capacity, minimum.

4.3.7 Transformer. - BUWEPS DWG 868863, or equivalent.

4.3.8 Special test equipment

4.3.8.1 Connector shorting strip. - The connector shorting strip shall be in accordance with figure 1, page 28 of NAVORD OD 12406.

4.3.8.2 Testing circuit. - The testing circuit shall be in accordance with figure 2, page 29 of NAVORD OD 12406.

4.4 Acceptance tests

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that Test Set Mk 339 Mod 0 complies with the requirements of 3.2 and 3.4.

4.4.2 Test conditions. - Unless otherwise specified, Test Set Mk 339 Mod 0 shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature. - Room ambient 65 degrees F to 95 degrees F. (18.33 degrees C to 35 degrees C)

4.4.2.2 Altitude. - Normal ground.

4.4.2.3 Vibration. - None.

4.4.2.4 Humidity. - Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.4.4 Test procedure

4.4.4.1 Dielectric. - The test specified in MIL-F-18670 Dielectric Test, shall be conducted in accordance with the instructions contained in NAVORD OD 12406, paragraph 4.

4.4.4.2 Insulation resistance. - The test specified in MIL-F-18870, Insulation resistance test, shall be conducted in accordance with the instructions contained in NAVORD OD 12406, paragraph 5.

4.4.4.3 Ground, damping signal, and power circuit continuity. - Test Set Mk 339 Mod 0, connected and tested as specified in paragraph 6 of NAVORD OD 12406, shall show the continuity and resistance values specified therein.

4.4.4.4 Phase Rotation lamp. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 7 of NAVORD OD 12406, the Phase Rotation lamps shall light as specified therein.

4.4.4.5 Signal Input dial. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 8 of NAVORD OD 12406, the AC voltmeter shall indicate the readings specified in 3.4.1.5 at the switch settings specified therein.

4.4.4.6 Output meter and Output Meter switch. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 9 of NAVORD OD 12406, the output meter shall indicate the meter readings specified in 3.4.1.6 at the switch settings specified therein.

4.4.4.7 Program Selector switch, decks D, E, F, and G. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 10 of NAVORD OD 12406, the Program Selector switch shall provide the voltages specified in 3.4.1.7.

4.4.4.8 Program Selector switch, deck A, and Signal Direction switch. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 11 of NAVORD OD 12406, the Program Selector switch, deck A, and the Signal Direction switch shall perform switching operations as specified therein.

4.4.4.9 Program Selector switch, decks B and C, and Phase Output lamps. When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 12 of NAVORD OD 12406, the Program Selector switch, decks B and C, and the Phase Output lamps shall operate as specified therein.

4.4.4.10 Tuning eye synchronism indicator. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 13 of NAVORD OD 12406, the tuning eye synchronism indicator shall operate at the voltages specified in 3.4.1.10.

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4.4.4.11 DC power supply output. - When Test Set Mk 339 Mod 0 is connected and tested as specified in paragraph 14 of NAVORD OD 12406, the DC power supply shall have an output voltage as specified in 3.4.1.11.

4.5 Pre-production and periodic production tests

4.5.1 Vibration

4.5.1.1 Vibration, transportation. - The Test Set Mk 339 Mod 0, packaged and packed as specified in section 5., shall be vibration tested as specified in Specification MIL-W-21927.

4.5.1.2 Vibration, operating condition

4.5.1.2.1 Orientation for vibration test. - The equipment shall be installed on vibration testing machines in such manner that the direction of vibration will be, in turn, along each of the three rectilinear orientation axes of the equipment. On a horizontal vibration testing machine, the equipment may be turned 90 degrees in the horizontal plane in order to vibrate it in each of the two horizontal orientations. The equipment may be installed in any position.

4.5.1.2.2 Vibration tests. - Each of the tests outlined herein shall be conducted separately in each of the three principal directions of vibration. All tests in one direction shall be completed before proceeding with tests in another direction. The equipment shall be secured to the vibration table as specified in 4.5.1.2.1, and shall be energized to perform its normal functions. All amplitudes of vibration specified are average amplitudes of the vibration table. If it is found necessary to repair any damage severe enough to constitute a failure of the instrument, the entire test shall be repeated unless otherwise directed by the bureau or agency concerned.

4.5.1.2.2.1 Variable frequency test. - The instrument shall be vibrated from 5 cps (or lowest attainable frequency) to 33 cps, in discrete frequency intervals of 1 cps, at the amplitudes shown in Table II. At each integral frequency, the vibration shall be maintained for five minutes.

4.5.1.2.2.2 Endurance test. - The instrument shall be vibrated for a total period of two hours, at the resonant frequencies chosen by the test engineer. If no resonance is observed, this test shall be performed at 33 cps. The amplitude of vibration shall be in accordance with Table II.

Table II

Frequency Range (cps)	Table Amplitude (inch)
5-15	0.030 ± 0.006
16-25	0.020 ± 0.004
26-33	0.010 ± 0.002

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4.5.2 Shock.

4.5.2.1 Shock, transportation, storage and handling. - The Test Set Mk 339 Mod 0, packaged and packed as specified in section 5, shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927.

4.5.2.2 Shock, operating condition. - Test Set Mk 339 Mod 0 shall be subjected to the following shock tests:

4.5.2.2.1 Method of mounting. - For all tests, the equipment shall be secured to the mounting bracket of the test machine as specified in MIL-S-901.

4.5.2.2.2 Shock test. - The test shall consist of nine blows being applied: three blows being applied parallel to each of the three principal axes of the equipment. The hammer shall drop from heights of one, three, and five feet. The weight of the hammer shall be 400 pounds. The shock testing machine shall comply with specification MIL-S-901.

4.5.3 Inclination. - Test Set Mk 339 Mod 0 shall be subjected to the following inclination tests;

4.5.3.1 The equipment shall be inclined at the rate of five to seven cycles per minute in one plane to angles of 45 degrees on either side of the vertical for a period sufficiently long to determine its electrical characteristics under such motion, or for a minimum of 30 minutes.

4.5.3.2 The test specified in 4.5.3.1 shall be repeated with the equipment reoriented 90 degrees to the plane in which it was originally tested.

4.5.3.3 At the conclusion of the tests specified in 4.5.3.1 and 4.5.3.2, the cyclic motion may be stopped and the inclination adjusted to any angle or angles within 45 degrees of the vertical for a sufficient time to investigate the electrical characteristics of the equipment.

4.5.4 Temperature and humidity. - Test Set Mk 339 Mod 0 shall be subjected to the following temperature and humidity tests:

4.5.4.1 Procedure I. - The instrument shall be placed in a test chamber where the temperature shall be between 20° and 38° C(68° and 100° F) with uncontrolled humidity. During the first two-hour period, the temperature shall be gradually raised to 71°C (160°F). The temperature of 71°C (160°F) shall be maintained during the next six-hour period. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20°C (68°F) and then raised to 38° C.(100° F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles). At the conclusion of the 240-hour period, the instrument shall be operated and the data recorded, and a visual examination shall be made for evidence of deterioration or corrosion. Distilled or

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demineralized water, having a pH value of between 6.5 and 7.5, at 25°C (77°F) shall be used to obtain the desired humidity. The velocity of the air throughout the test area shall not exceed 150 feet per minute.

4.5.5 Fungus resistance test

4.5.5.1 Organisms. - Immediately after Test Set Mk 339 Mod 0 has been subjected to the temperature and humidity test as specified in 4.5.4.1; it shall be subjected to the fungi tests as specified in paragraphs 4.4.3.1 - 4.4.3.3 of MIL-T-945.

4.5.6 Marine environment test. - Test Set Mk 339 Mod 0 shall be subjected to the simulated sea atmosphere test under conditions specified in section 4 of MIL-T-18404. At the conclusion of the test, Test Set Mk 339 Mod 0 shall meet the performance requirements of this specification.

4.5.7 Operating life. - Test Set Mk 339 Mod 0 shall be subjected to the following accelerated life test.

4.5.7.1 Normal operation. - Test Set Mk 339 Mod 0 shall be operated after a 20 minute warm up period in a room with an ambient temperature of 25 plus or minus 5 degrees C and 50 plus or minus 3 percent relative humidity, and critical performance parameters shall be measured and logged as "reference test data" for comparison with subsequent tests.

4.5.7.2 Temperature and altitude tests. - After performance parameters have been established, Test Set Mk 339 Mod 0 shall be placed in a temperature - altitude chamber and tests conducted as specified in paragraphs 4.4.4. through 4.4.4.20 of MIL-T-945. Failure of the instrument to pass the tests as specified will be cause for rejection.

4.6 Waiver of tests. - Tests specified in 4.5.1.1 and 4.5.2.1 may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Function tests. - At the conclusion of the tests specified in 4.5.1.1 and 4.5.2.1, functional and other tests, as applicable, shall be conducted on the test set to determine freedom from operational malfunction and compliance with all the requirements of section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - In addition to the following Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning. - The exterior surfaces only of the test set shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - The exterior surfaces only of the test set shall be dried in accordance with procedure D-1 or D-4 of Specification MIL-P-116.

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5.1.1.3 Preservative application. - None required.

5.1.1.4 Unit packaging. - The test set shall be individually packaged in accordance with BUWEPS DWG 1349424.

5.1.2 Level B. - Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying, and preservative application. - The cleaning, drying, and preservation of the test set shall be in accordance with paragraph 5.1.

5.1.3.2 Unit packaging. - The test set shall be individually packaged in accordance with the manufacturer's commercial practice, when such practices have been proven by test to conform to all applicable requirements. The packaging shall be adequate to provide protection from physical damage, corrosion or other forms of deterioration during direct shipment from supply source to the first receiving activity for immediate use. The method used shall be commensurate with shipping at the lowest practical commercial rate.

5.2 Packing

5.2.1 Level A. - The test set, packaged as specified in paragraph 5.1, shall be packed in accordance with BUWEPS DWG 1349424.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - The test set, packaged as specified in paragraph 5.1, shall be individually packed in a manner to insure carrier acceptance for safe delivery at destination at the lowest rate. The shipping container shall conform with the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal marking. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129, including the serial number of the test set contained therein.

5.3.2 Special marking. - The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

6.1 Intended use. - Test Set Mk 339 Mod 0 is intended for use as test equipment in the service and maintenance of all mods of Amplifier Mk 140.

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6.2 Ordering data. - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification, applicable drawings, selection of applicable levels of preservation, packaging, and packing, and other documents.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to paragraph 4 of this document.

6.3 Preparation for delivery criteria. - Criteria for use of proper level of preservation, packaging and packing shall be as follows:

6.3.1 For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.3.2 For Level B. - This level shall be used only for those items which are to be subjected to multiple domestic shipments, handling and stored in warehouses or other structures affording equivalent protection from weather.

6.3.3 For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Packing and packaging for storage. - Packing and packaging for storage shall comply with the requirements specified in 5.1 and 5.2.

6.5 General safety precautions. - A sign listing operating instructions for Test Set Mk 339 Mod 0 shall be prepared in accordance with instructions contained in OP 2555.

NOTICE: When Government drawings, specification, or other data are used for any purpose other than in connection with a definitely related Governmental procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 48-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*:

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT [ ] OR A SUBCONTRACT [ ]

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
[ ] YES [ ] NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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Chief, Bureau of Naval Weapons (RREN-5)

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